

HRISHIKESH BELAGALI

Undergraduate Student

📍 East Lansing

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SUMMARY

Undergraduate CS and Math double major interested in quantum algorithms, high performance computing, molecular dynamics simulations, Monte Carlo methods and other optimization techniques

SKILLS

Languages: Python, $\LaTeX$, JavaScript, C++

Packages: Qiskit, NumPy, PyTorch, Matplotlib

Softwares: LAMMPS, OVITO, NAMD, VMD

EDUCATION

Computer Science (B.S)

Expected Graduation: 2027

Current GPA: 4.0

Michigan State University

CONFERENCES

2020

IISc, Bengaluru

12th Biennial Lake Conference

Presented a case study on the destruction and conservation efforts of Sarakki Lake

Secured 3rd place and awarded Sahyadri Young Ecologist of the Year

EXPERIENCE

2024-

PRL, MSU

Vermaas Lab

Explored thylakoid membrane protein interactions employing molecular dynamics simulations using NAMD and VMD.

PROJECTS

Python

Qiskit

Discretized Quantum Adiabatic Evolution

[github.com \(private\)](#)

Implemented a quantum adiabatic algorithm to solve QUBO type problems using the Trotter-Suzuki approximation and quantum Ising model formulation techniques

Python

Qiskit

Quantum Arithmetic Circuits

[github.com](#)

Implemented quantum ALUs (Arithmetic Logic Units) for elementary operations augmented by an approximate Quantum Fourier Transform (AQFT) to improve accuracy in the presence of decoherence.

Python

OVITO

pyMOL

Genetic Annealing to Determine Protein Structures

[github.com](#)

Hybrid genetic annealing algorithm to determine protein structures through the optimization of Irbäck's off-lattice model energy equation ($\text{RMSD} < 3.0$). Use of molecular dynamics to simulate protein folding.

Python

Quadratic Unconstrained Binary Optimization

[github.com](#)

Stochastic tunneling-enhanced simulated annealing for solving QUBO-type problems.

Python

Monte Carlo Integration

[github.com](#)

Monte Carlo integration with importance sampling to numerically evaluate complex integrals.

Python

Langevin Monte Carlo

[github.com](#)

Metropolis-adjusted Langevin algorithm for sampling from intractable probability distributions.

Python

Lattice Grid Optimization

[github.com](#)

Designed and implemented a simulated annealing algorithm to visualize how cooling schedules affect clustering in a two-color lattice grid.

Python

RESTful API

AP Survey Automation and Analysis

[github.com](#)

Created a Google Forms API script to generate over 30 AP exam surveys, collecting 1,000+ responses. Led a team to analyze survey data, identifying popular resources for 50,000+ AP students.